Thanh Vuong

Data Scientist / Machine Learning Engineer

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Education

Master of Artificial Intelligence, RMIT University 2024 - 2026
Grade: Distinction | Scholarship Awarded

- **Best Project Award** – AI Postgraduate Project: *"GuideFlow: An Explainable, LLM-Constructed Clinical Decision Support System for Diabetes Management"* (industry partner: AstraZeneca).

Bachelor of Engineering, Vaasa University of Applied Sciences 2014 - 2018
Software Engineering

Work Experience

Be Group JSC - Ride-hailing Application 08/2022 – 09/2026
Senior Data Scientist/Machine Learning Engineer Lead

Led a team of 4 ML engineers to design and maintain real-time ML models for pricing, dispatch, and recommendations, car pool and delivery systems.
Developed and optimized production ML pipelines, improving scalability and efficiency.

Internal AI Agent Platform

- Designed and built an **internal multi-agent AI platform** giving the DS & engineering teams a unified assistant for daily knowledge work.
- Use cases: Q&A over **Confluence docs, GitLab code, and BigQuery datasets**; dispatch rule configuration & explanation; on-call troubleshooting and operational support.
- Built an **internal MCP server** bridging agents to Confluence, GitLab, and BigQuery, so any agent can answer with grounded company knowledge.
- Delivered a **Dispatch configuration UI** on top: business team configures dispatch scoring rules via web UI; AI explains rule impact (trace driver, simulate, estimate). Rule changes **hot-reload into the production dispatch engine** in real time, replacing ad-hoc code-edit-and-deploy workflows.

Dynamic Pricing Engine

- Designed and implemented a surge pricing model, increasing revenue by **70-100%** during peak hours and rainy conditions.
- Developed a demand-supply forecasting model to predict market conditions for the next 30 minutes (6 time steps).
- Built a pricing engine that dynamically adjusts surge factors based on predicted demand and supply, optimizing balance in future time steps.

Dispatch Engine

- Built an **A/B testing framework** to validate dispatch strategies and model improvements in real-time production environments.
- Developed an enhanced **driver–rider scoring model** that increased **completion rate (CR) by an additional 6%** compared to the baseline model.
- Implemented a **Multi-Armed Bandit (MAB) testing module** to dynamically allocate traffic across dispatch policies, accelerating learning under changing market conditions.
- Implemented a linear sum assignment algorithm to maximize overall rider-driver match scores. Increased **completion rate (CR) by 10%** and reduced **average pickup distance by 30%**.

Food Recommendation Systems

- Developed a hybrid recommendation model combining content-based filtering and collaborative filtering.
- Implemented a DeepFM model, improving ranking **NDCG by 20%** and achieving a **15%** increase in CTR through A/B testing.
- Recommendation engine contributes **20%** of food product traffic revenue.
- Applied byte-pair encoding for Vietnamese tokenization in the food domain.
- Designed a keyword extraction model that analyzes user interactions to recommend relevant dishes.

Be Search Engine - Food and Map Search

- Architected the search systems for **beMap** and **beFood**.
- Designed search logic optimized for business objectives, ranking items based on rating, partnerships, distance, and relevance.
- Developed a keyword suggestion feature using Elasticsearch, ranking terms dynamically via exponential moving average (EMA).
- Search system drives **60%** of total food product traffic revenue.
- Enabled real-time data synchronization.

Waiting Time Estimation Model

- Built a predictive model to estimate driver waiting times within geofenced areas.
- Developed a demand forecasting model to generate real-time heatmaps for dynamic driver allocation.

Food Delivery ETA Model

- Designed a predictive model for estimating delivery time ranges, enhancing customer experience and reducing driver idle time.
- Improved **on-time delivery rate by 50%** compared to baseline statistical methods.

Dropoff Recommendation System

- Developed a model predicting user drop-off locations based on historical interactions, including location, time of day, and day of the week.
- Enhanced user experience, with **80%** of users adopting the feature.

Sen Do Technology JSC - E-commerce Platform 07/2019 – 05/2022

Data Scientist

Brand Detection

- Developed **Computer Vision AI models** to extract brands from product images.
- Implemented **Faster R-CNN, ResNet, and OCR models** to detect and classify brands into 500 categories.
- Achieved **over 90% accuracy**, improving market management efficiency by **300%**.

Sendo Search Engine Optimization

Led efforts to enhance the Sendo search engine, designing ranking logic and optimizing product visibility.
*Result: CTR improved from **38% to 47%**.*

Search Query Category Mapping

- Developed a **deep learning model** to categorize search queries, improving product relevance.
- Achieved **97% precision and 80% recall**, increasing overall search CTR by **1%**.

Search Query Auto-Correction

- Implemented a **misspelling correction model** that refines user queries before processing in Elasticsearch.
- Based on research paper: *Using the Web for Language-Independent Spellchecking and Autocorrection (Google)*.
- Improved query accuracy by **60%** from test datasets.

Search Query Auto-Filtering

- Developed a **sequence labeling model (CRF)** to extract product attributes (price, size, color, location) from queries.
- Automatically applies relevant filters to refine search results.

Synonym Dictionary for Elasticsearch

- Built a **dictionary of 2,000+ synonym pairs** to improve query interpretation.
- Trained a **Word2Vec skip-gram model** on 19 million product descriptions.
- Extracted **highly similar words** to enhance Elasticsearch search accuracy.

Product Title Scoring System

- Developed a **scoring model** to rank product titles based on relevance and quality.
- Implemented **two distinct models**:
  - **Title Spamming Score**: Detects and penalizes spam-like product titles.
  - **Title Completeness Score**: Rewards well-structured, informative titles.
- Titles flagged as spam are **excluded from top search results**, improving user experience.

FPT Software 07/2018 – 07/2019

Machine Learning Engineer

OCR-Based Vehicle Registration Recognition

- Developed an OCR-based model for vehicle registration recognition, achieving **90% accuracy**.
- Improved document processing efficiency through automated extraction of license plate details.

Anomaly Detection in Automotive Sensor Data

- Implemented an anomaly detection system to identify irregularities in automotive sensor data.
- Enhanced predictive maintenance by detecting potential failures before they occur.

Founding & Entrepreneurship

MasterHive.AI — Health-Data Platform Startup 2026 – Present

Founding Member

Built a health-data platform from first design to a live system serving a province-scale user base, spanning three products.

MasterHive — Electronic Health Record Platform

- In production across a provincial network of **hospitals, district health centers and commune clinics**. Clinicians create, sign and manage records through configurable clinical forms, with each patient's history aggregated across facilities.

MasterFlow — Provincial Health Database Platform

- Ingests and standardizes health-examination records from every facility in a province, enforces **data-quality validation**, and routes them through approval; serves as the province's **single reporting channel** to the national Ministry of Health.
- In its first production deployment inside a **government data center** on a **.gov.vn** domain, with a partner API opening for the city app.

MasterCare — Patient Mobile App (iOS & Android)

- Patients browse their medical history as a timeline, get **AI-generated health summaries with citations** back to source records, and receive medication and follow-up reminders.

Volunteer Projects

Nhu Chua He Co Cuoc Chia Ly – People Reunification Portal 2024 – Present
Volunteer Data Scientist / AI Engineer

- Developed a **GenAI-powered matching system** to support locating missing persons and reconnecting separated families.
- Designed AI workflows leveraging **NLP and semantic search** to analyze incomplete, noisy, and unstructured user-submitted information.
- Collaborated with **FPT Cloud** to deploy and operate production systems at scale.
- Successfully launched public platforms: nhuchuahecocuocchialy.com and portal.nhuchuahecocuocchialy.com.
- Built a secure user portal for case submission, authentication, and AI-assisted candidate suggestion.

Other Projects

AI Pharmacist Assistant — Major Vietnamese Pharmacy Retail Chain 2026
AI / ML Architect — Proof of Concept

- POC for an in-store AI assistant supporting pharmacists across a 1,000+ store retail chain: contextual customer engagement (Customer 360), cross-sell / up-sell, smart FEFO alerts, and omni-knowledge helpdesk (drug info, SOPs, HR).
- Designed a multi-agent architecture (Intent Router, Context Manager, specialized agent swarms) delivered as a POS-integrated mini-app with A/B testing for safe rollout.

Certifications and Awards

Best Project Award - RMIT AI Postgraduate Project 2026
GuideFlow - AstraZeneca partner
Excellent Project Award - Dispatch Engine 2025
The Contributor Award - Be Group 2024
Best Employee Recognition
Top Performer of The Year - Data Scientist at Be Group 2023
Top Performer of The Year - Data Scientist at Be Group 2022
Best Employee Of Organic Growth Tribe - Data Scientist at Sendo 2021
Best Employee Of The Year Award - Data Scientist at Sendo 2020
Data Science Mentor at MASSP - Math and Science Summer Program 2021

Technical Skills

LLM & AI Agents: LangGraph, LangChain, LlamaIndex, multi-agent systems, MCP (Model Context Protocol), tool calling, OpenAI / Anthropic (Claude) APIs
RAG & Vector Search: RAG pipelines, embeddings, hybrid & semantic search, grounded answers with citations; pgvector, Qdrant, Milvus, Elasticsearch
LLM Evaluation & Observability: DeepEval, Langfuse, LLM-as-a-judge, RAG & agent evaluation
Machine Learning & Deep Learning: PyTorch, TensorFlow / Keras, scikit-learn, LightGBM, XGBoost, CatBoost; Time Series, NLP, Recommender Systems, Computer Vision
MLOps & Deployment: MLflow, ONNX, A/B testing & Multi-Armed Bandit, Docker, Kubernetes, CI/CD, FastAPI, Grafana
Programming: Python, SQL
Big Data Technologies: Apache Spark, Kafka, Hadoop, Airflow, GCP, BigQuery, Pub/Sub
Databases: PostgreSQL, MongoDB, Redis, Elasticsearch

Languages

Vietnamese: Native
English: Professional working proficiency

References

- Assoc. Prof. Xuan-Hieu Phan, Head of Research and Development Center - Sendo Technology JSC
Email: pxhieu@gmail.com
- PhD. Hoang-Huy Nguyen - AI Director at Techcombank
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